

Cascade Module (Plug-and-Play Spec)

Adapter Contract

- `SeedAdapter(seed_input, host_context) -> SeedRegion`
- `ConstraintAdapter(host_context) -> Constraints`
- `ScoreAdapter(host_context) -> Score(x) in [0,1]`
- `ParamsAdapter(host_context) -> {theta_activate, theta_expand, theta_continue, theta_bind, Rmax, ...}`
- `DistanceMetric(host_context, SeedRegion) -> Dist`
- `Activate(signal, theta_activate) -> {0,1}`
- `Expand(signal, SeedRegion, Dist, theta_expand, Params) -> Candidates`
- `Filter(Candidates, Constraints, Params) -> Viable`
- `SelectBest(Viable, Score) -> x`
- `Refine(Viable, r, signal, Constraints, Params) -> Viable`
- `CommitAdapter(x_star, host_context) -> {commit_kind, commit_payload}`

Reference pseudocode

```
function Cascade_Run(signal, seed_input, host_context):
SeedRegion = SeedAdapter(seed_input, host_context)
Constraints = ConstraintAdapter(host_context)
Score = ScoreAdapter(host_context)
Params = ParamsAdapter(host_context)
Dist = DistanceMetric(host_context, SeedRegion)

active = Activate(signal, Params.theta_activate)
if active == 0:
return {status: "inactive", x_star: null, commit_kind: null, commit_payload: null}

Candidates = Expand(signal, SeedRegion, Dist, Params.theta_expand, Params)
Viable = Filter(Candidates, Constraints, Params)

for r in 1..Params.Rmax:
x_r = SelectBest(Viable, Score)
if Score(x_r) >= Params.theta_continue:
break
Viable = Refine(Viable, r, signal, Constraints, Params)

x_star = SelectBest(Viable, Score)
if Score(x_star) >= Params.theta_bind:
{kind, payload} = CommitAdapter(x_star, host_context)
return {status: "commit", x_star: x_star, commit_kind: kind, commit_payload: payload}
else:
return {status: "no_commit", x_star: null, commit_kind: null, commit_payload: null}
```